

Assessment 3

Cloud Computing

Fundamentals

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1. Introduction

This assessment will explore and demonstrate the utilization of Amazon Web Services (AWS). The purpose of this research is to gain a deeper understanding of AWS's features, security concepts, virtualization, and resource pooling in a cloud environment. Among many cloud providers, AWS has been chosen as our preferred cloud service provider due to its excellence in overall service variety. Cloud computing offers scalable and on-demand access to IT resources, enabling efficient virtualization and resource pooling, which are essential for modern application deployment (Marinescu, 2021). While Azure excels as a suite of Microsoft products, and Google Cloud Platform stands out in AI and data analytics, AWS is the ideal choice for this deployment scenario because it aligns with industry standards for cloud-based application deployment.

When discussing budgeting, scalability, maintenance, and management, cloud computing has become a solution for businesses nowadays. Unlike traditional IT infrastructure, which requires businesses to pay a significant amount for upfront investments and maintenance, cloud computing offers several advantages. Compared to traditional IT infrastructure, users only need to pay for the resources they use, with high scalability and availability. This benefit allows businesses to adapt quickly to market demands.

This project explores the implementation of AWS as a leading cloud platform. The combination of three deployment models is involved: Infrastructure as a Service (IaaS), such as EC2 and S3, where virtual machines and storage are provisioned; Platform as a Service (PaaS), which supports deployment without managing hardware; and Software as a Service (SaaS), where applications are accessed via the web without installation. The deployment model used in this project is public cloud, which is managed by AWS for cost efficiency and scalable solutions. These advantages make cloud platforms particularly suitable for educational environments and web application deployment projects, including open-source platforms like Moodle (Buyya et al., 2021). Public cloud also allows users to experience the full capabilities of AWS, ensuring secure and virtualized access to computing resources

2. AWS as The Chosen Service Provider

For this project, AWS was selected because we learned about it in class. Besides, AWS is an ideal platform for deploying and managing cloud-based applications with wide range of services, and high reliability. As a pioneer in cloud computing, AWS provides flexible, scalable services and high standard documented solutions to support computing, storage, networking and security cases. This project will describe the implementing of AWS tools and features to support the deployment of Moodle web application with low cost and high performance. According to Forgeahead (2024), one key benefit of using

AWS is its elasticity, which allows businesses to automatically scale resources up or down based on real-time demand, ensuring optimal performance and cost-efficiency.

3. Deployment Model

The deployment model used in this project is the public cloud deployment model, which is well-suited for educational platforms like Moodle. This is demonstrated by the use of Amazon Web Services (AWS) EC2 instances within the AWS Academy Learner Lab. Public cloud deployment is preferred for educational applications due to its flexibility, reduced infrastructure costs, and the ability to quickly scale based on user demand (Hussain et al., 2021).

In this project, the EC2 infrastructure applied with VPC to make it efficient and high scalable for deploying cloud web-based learning management systems. Besides by utilizing AWS EC2, the project benefits from a virtual server that provides flexible computing capacity in the cloud. EC2 instances allow for easy resource scaling, enabling the Moodle server to handle varying workloads and user demands without service interruptions. The public cloud model supports cost-effective deployment because it eliminates the need for upfront hardware investments and ongoing physical maintenance, which is crucial for academic institutions with limited IT resources.

Furthermore, the VPC ensures that the deployed application is hosted within a logically isolated network environment. VPC offering control over networking settings such as IP address ranges, and 2 subnets which are public for the web server and private for the database. This contributes to the security and reliability of the Moodle platform, even in a shared infrastructure.

The use of Elastic IPs ensures consistent accessibility, and security groups control network access. With implementing public cloud deployment in AWS environment, it helps for the future research, as the infrastructure can easily accommodate more users, advanced analytics, or integration with other cloud-native services.

4. Deployment Procedures

4.1 Create VPC

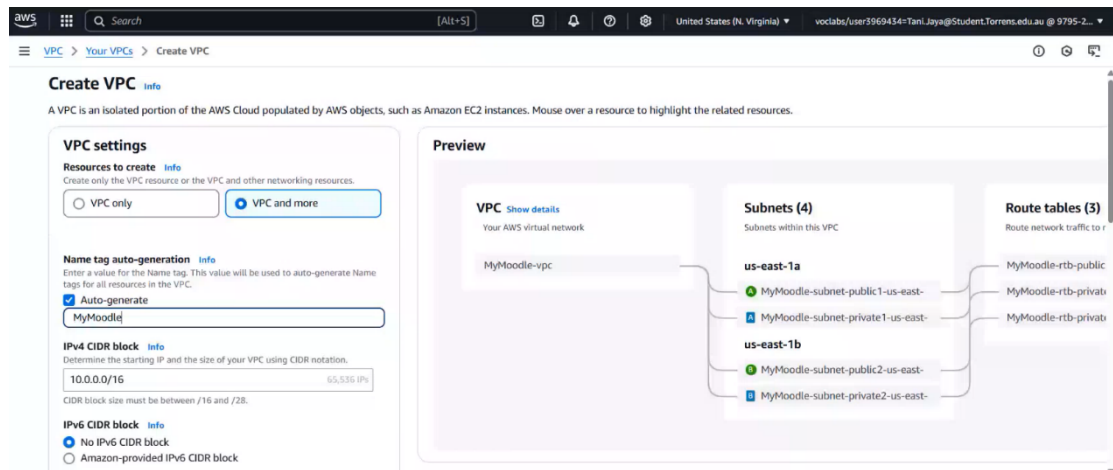


Figure 1 Create VPC

The first step on this deployment was setup the VPC (Virtual Private Cloud). VPC as the fundamental building block for the networking in AWS. Creating a VPC (Virtual Private Cloud) allows users to launch the EC2 instances in secure network and has control IP range, subnets, and the availability zones. Those features ensure better control, and protection of Moodle deployment environment.

4.2 Create Security group

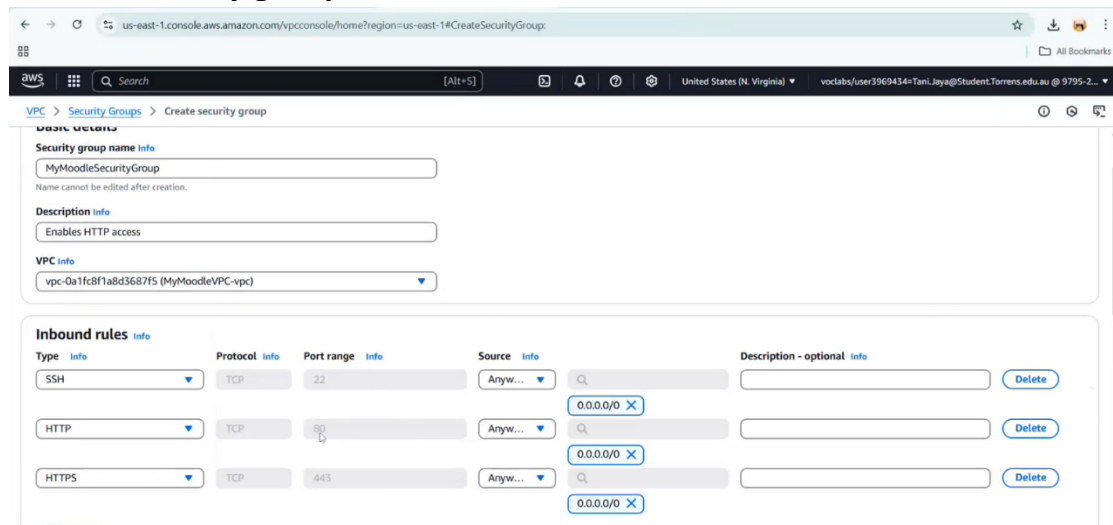


Figure 2 Create Security Group

The second step created the security group. Security group is important to give the control access to the web server and database. In this section there were 3 inbound rules which assigned: SSH, HTTP, and HTTPS. Security groups restrict traffic by IP and port, while IAM policies define user and service permissions. AWS also offers automated backups and SSL encryption for data in transit, ensuring confidentiality and integrity (Fernandez et al., 2020).

4.3 Create new instance

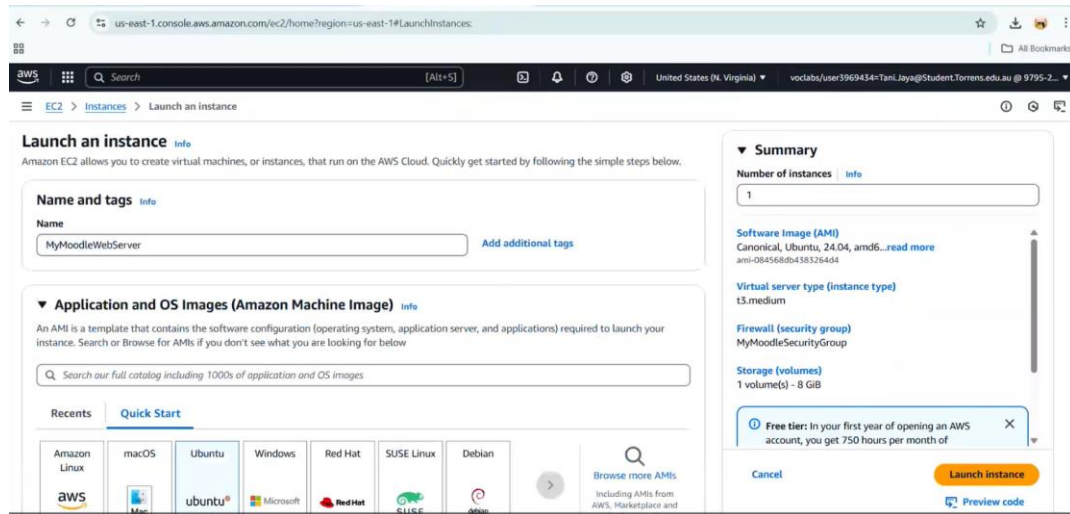


Figure 3 Create new instance

On this step, new instance was created as MyMoodleWebServer, with ubuntu as the operating system. There are several reasons to choose Ubuntu for this project. One of the reasons is Ubuntu and AWS offer free services for the beginners, with cost effectivity, easy controlling and managing EC2 instance, and they provided the well documented with online community support. The t3.medium for the instance type was chosen because it has 2 vCPUs and 4 GB RAM which support a lightweight production-ready like Moodle and allowing Moodle to handle high workload without high cost, making ideal for an education purpose. Ubuntu is commonly chosen for Moodle deployment due to its compatibility with PHP-based applications and its ease of package management in cloud environments (Nadarajan et al., 2021).

4.4 Select existing security group on the create new instance setup

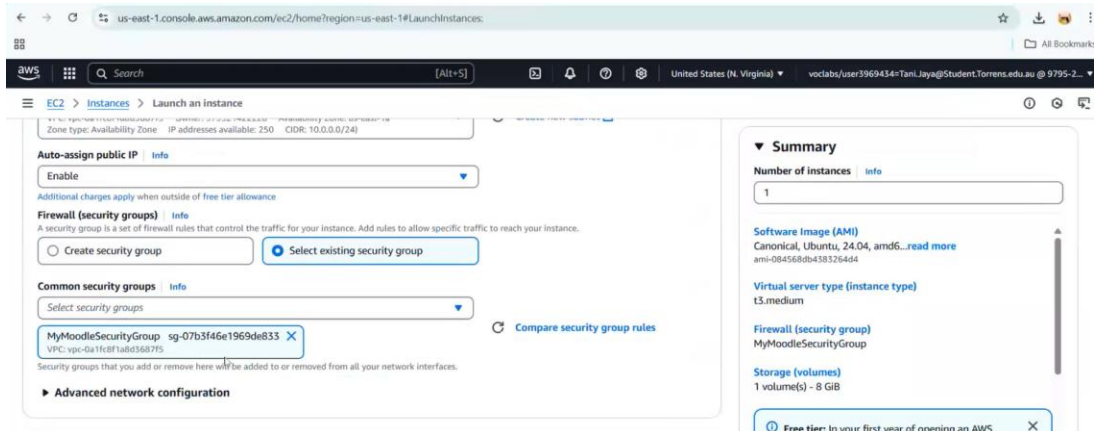


Figure 4 Select existing security group

Attaching existing security group which already made before making this instance more secure, reduce configuration error and ensure consistency in AWS environments like AWS Learner Lab where custom security configuration may be restricted.

4.5 Associate Elastic IP Address

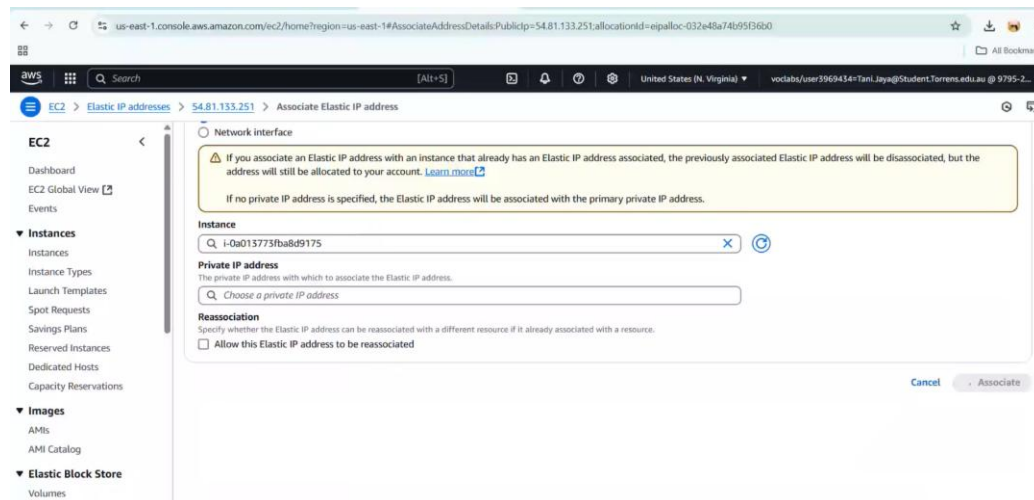


Figure 5 Associate Elastic IP Address

In this project provides with Elastic IP to ensure that the Moodle web server has a static public IP address which doesn't change even if the EC2 instance is stopped or restarted. Elastic IP addresses provide a fixed endpoint in dynamic cloud environments, enabling reliable application access across instance lifecycle events (Patel & Desai, 2023).

4.6 Installation Process

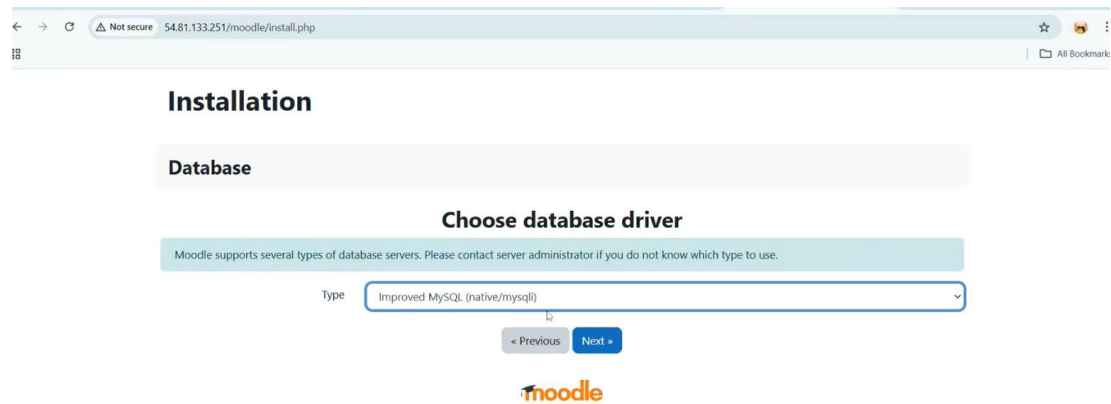


Figure 6 Choosing MySQL in installation process

In the installation process, MySQL has chosen for Moodle web application. Since MySQL is widely supported, open-source relational database that is fully compatible with Moodle's backend requirements. MySQL is recommended for Moodle due to its proven compatibility, scalability, and performance in handling educational content and user data (Moodle Docs, 2024).

4.7 Finish the Installation

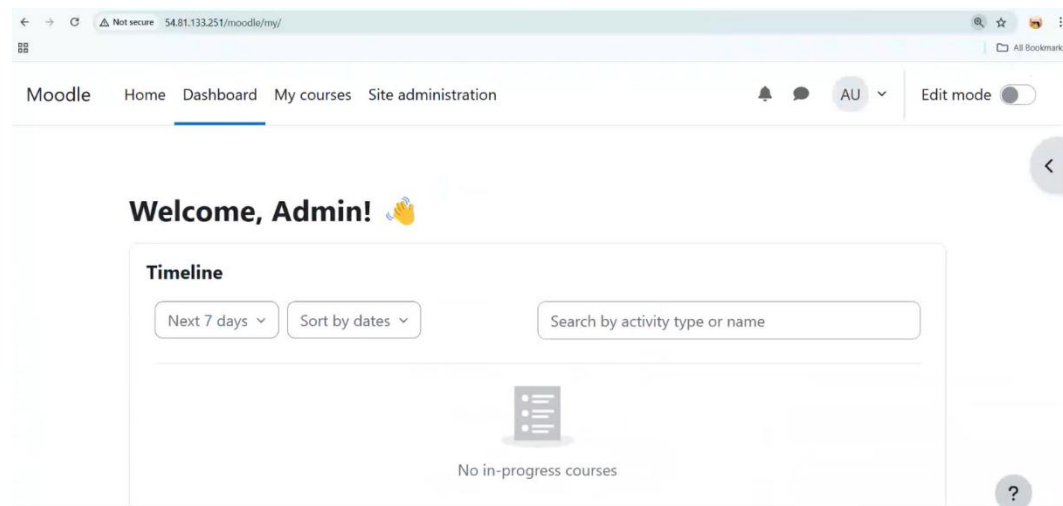


Figure 7 Finish the Installation

At the end of the installation process, Moodle was successfully deployed and configured to match the specified user profile and system requirements. All necessary services, including the web server, database, and Moodle application files, were properly set up and integrated. The platform is now accessible via a public IP

address, allowing users to interact with the learning management system at the following URL: <http://54.81.133.251/moodle/>.

5. Analysis The Application

This project could be developed for the further purpose. Since the Moodle application was deployed on AWS which provides wide services to support the robustness, scalability and the security of the application. Below are some points to be developed for this project:

- 1) Implementing Application Load Balancer to allow the system to handle the high workload without downtime.
- 2) Integrating Amazon RDS for the database management for better performance, and automated backups.
- 3) Enabling HTTPS via SSL certificates to ensures encrypted communication between users and server.
- 4) Enabling AWS CloudWatch to alert the administrator from threat activity.

6. Conclusion

In the conclusion, this project documented how to implement a public cloud infrastructure to deploy a web application which is Moodle. Cloud-hosted Moodle supports high scalability and consistent performance under growing user demand (Chowdhury & Ahmed, 2023). With using Amazon web services (AWS) as an open-source learning management system, students provided an account to manage, control, deploy and install which implements some procedures starting with create a secure virtual network (VPC), configure the security group, using the elastic IP address and build an EC2 instance. All these configurations adopted a public cloud deployment model to deliver the needs of cost efficiency, accessibility, and easy maintainability of infrastructure, which is relevant for academic and research purposes.

To ensure the reliable performance, the project also highlighted the significance of network configuration, system security, and automated services. Some of cloud practices were used during the setup, including browser-based SSH for deployment in restricted settings like AWS Academy Learner Lab, role-based access, and structured resource management. The further investigation is recommended in areas such as load balancing, automated backups, and high-availability architecture to prepare for production-scale use. This research contributes to a deeper understanding of deploying web-based applications in the cloud and provides the foundation for future enhancements in education technology infrastructure. Cloud computing enhances e-learning platforms by offering flexibility and cost efficiency (Yusop et al., 2022).

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